

Recap I

Observe: complex STFT coefficient X_{rn} and its power $Y_{rn} = |X_{rn}|^2$

Model:

$$X_{rn} \sim \text{i.i.d. } \mathcal{CN}(0, V_{rn}), \quad V_{rn} = \sum_{k \in \mathcal{K}} a_{rk} e^{-\lambda_k \tau_n}.$$

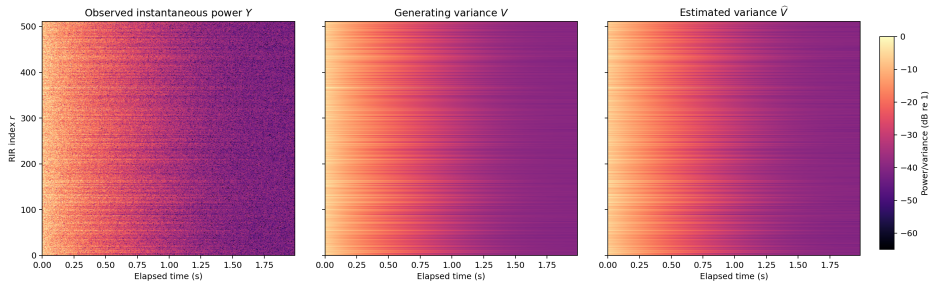
In latent variable form,

$$X_{rn} = \sum_{k \in \mathcal{K}} Z_{rnk}, \quad Z_{rnk} \sim \text{i.i.d. } \mathcal{CN}(0, v_{rnk}), \quad v_{rnk} = a_{rk} e^{-\lambda_k \tau_n}.$$

Loss: negative log-likelihood

$$\mathcal{L} = \sum_{rn} \left(\log V_{rn} + \frac{Y_{rn}}{V_{rn}} \right) \sim d_{\text{IS}}(Y|V).$$

Recap II



Recap III

Wiener-ratio of slope k , at frame n in RIR r ,

$$\rho_{rnk} = \frac{v_{rnk}}{V_{rn}}.$$

Posterior energy $\mathbb{E}[|Z_{rnk}|^2 \mid X_{rn}]$ is given by

$$S_{rnk} = \left(\frac{v_{rnk}}{V_{rn}} \right)^2 Y_{rn} + \left(1 - \frac{v_{rnk}}{V_{rn}} \right) v_{rnk}.$$

From this we build the surrogate loss function

$$\mathcal{Q}_k = \sum_{rn} \left(\log v_{rnk} + \frac{S_{rnk}}{v_{rnk}} \right),$$

which gives the close form solution for a_{rk} , which further makes the update of λ_k a convex optimization problem - **SAGE**.

Posterior energy

Rewrite

$$S_{rnk} = \left(\frac{v_{rnk}}{V_{rn}} \right)^2 Y_{rn} + \left(1 - \frac{v_{rnk}}{V_{rn}} \right) v_{rnk}$$

to

$$\begin{aligned} S_{rnk} &= \underbrace{v_{rnk}}_{\text{prior}} + \underbrace{\left(\frac{v_{rnk}}{V_{rn}} \right)^2 (Y_{rn} - V_{rn})}_{\text{posterior update}} \\ &= v_{rnk} + \rho_{rnk}^2 \epsilon_{rn}. \end{aligned}$$

Only the part where ρ_{rnk} is large will contribute to loss, which in turn help move λ_k to the right direction.

Contribution-weighted SAGE

Adding weights to the surrogate loss,

$$\mathcal{Q}_k = \sum_{rn} \rho_{rn}^p \left(\log v_{rnk} + \frac{s_{rnk}}{v_{rnk}} \right).$$

This really works!

Convergence I

Setup

- ▶ Batch decay update over one frequency bin with $K = 2$ slope components and $R = 512$ RIRs with amplitudes sampled from joint Gaussian distribution.
- ▶ Both component T_{60} values are initialized at 2.00 s.
- ▶ The following pages show the T_{60} trajectories for three runs with $p = 0, 1, 2$.
- ▶ Convergence criteria: $|\lambda_k^{(i)} - \lambda_k^{(i-1)}| < 10^{-6}$ for all k .
- ▶ No strict correspondence between the surrogate loss and the global loss.

Convergence II

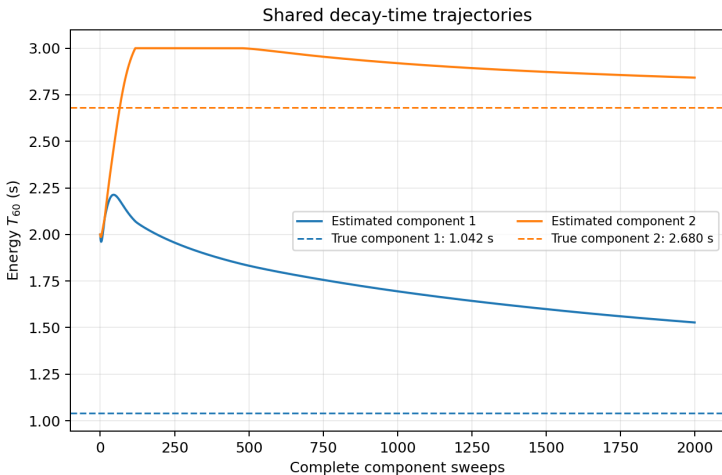


Figure: Native SAGE, - no weight in surrogate loss $p = 0$.

Convergence III

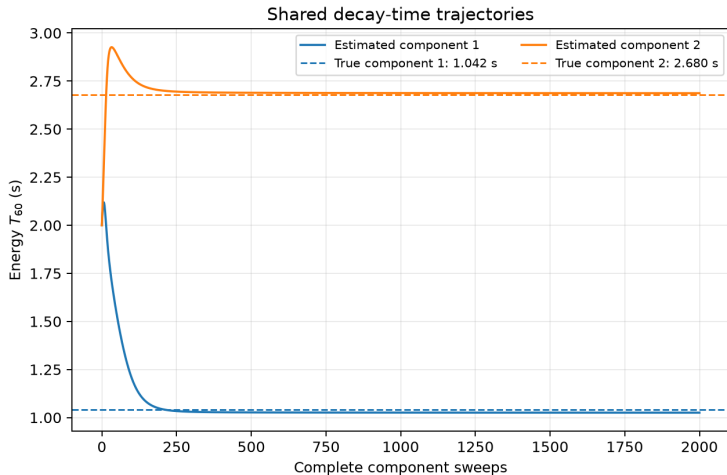


Figure: CW-SAGE with $p = 1$.

Convergence IV

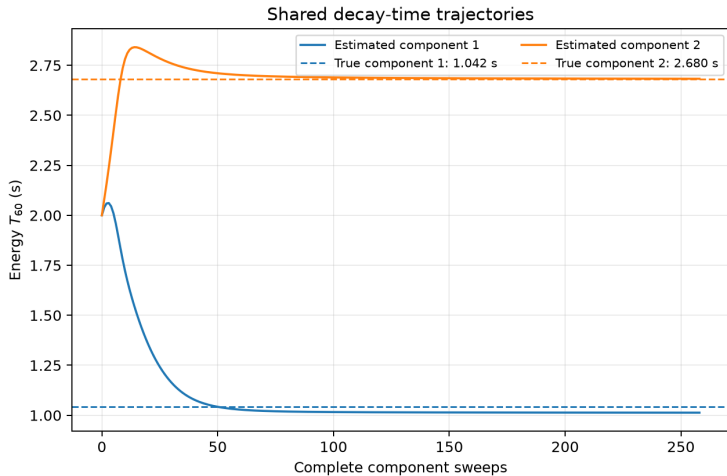


Figure: CW-SAGE with $p = 2$. Notice the bigger bias!

Robustness to initialization I

Setup

- ▶ Same data as previous setup: $T_{60,1} = 1.042$ s, $T_{60,2} = 2.680$ s
- ▶ Initialise with T_{60} pairs within $[0.50, 3.00]$ s.

Robustness to initialization II

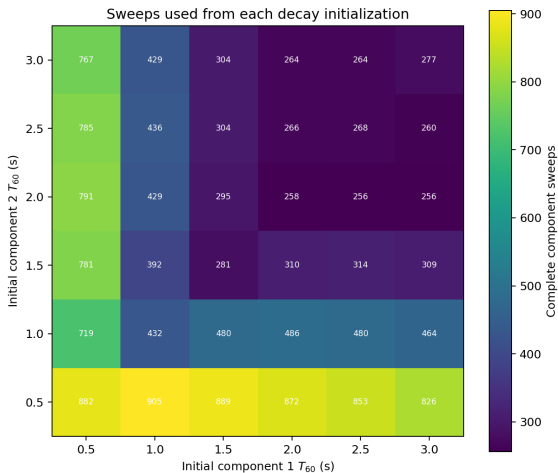


Figure: Sweeps needed for convergence at different initializations.

Robustness to initialization III

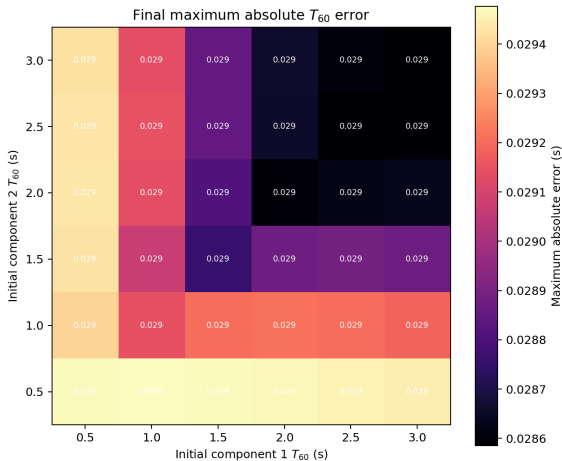


Figure: Decay sensitivity to initialization.

Robustness to initialization IV

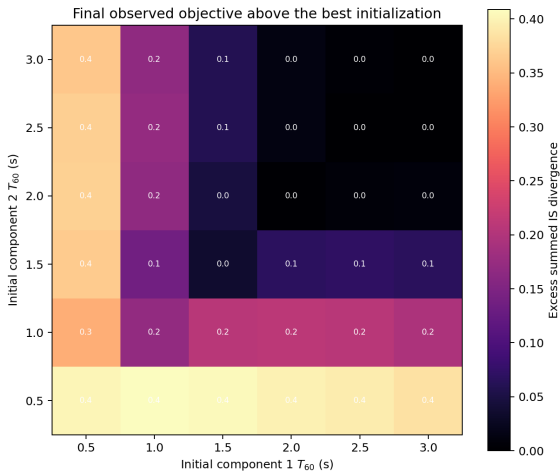


Figure: Global loss is also compromised.

Robustness to true ΔT_{60} I

Setup

- ▶ For arbitrary frequency bins $F = 100$, generate $R = 512$ RIRs with two slope components with T_{60} values sampled from Dirichlet distribution.

Robustness to true ΔT_{60} II

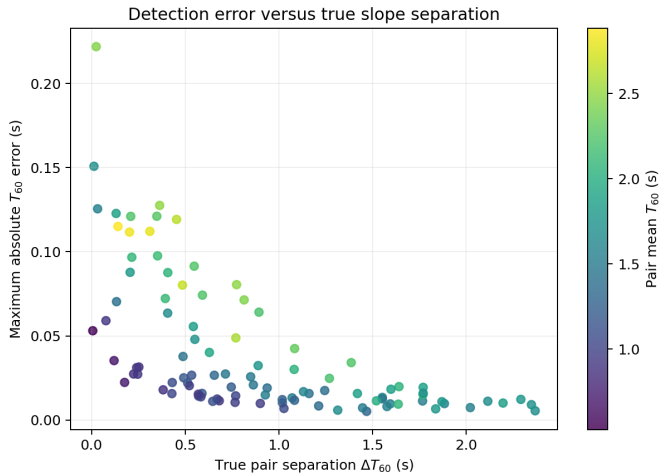


Figure: Decay sensitivity to true ΔT_{60} in synthetic data.